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The formation of Si/B-bearing innerlayer and its effect on the growth and adherence of chromia scale formed by pre-oxidation of cobalt-chromium based alloy

Chun Yu, Liangzhen Zhu, Yanjin Lu, Yi Zhao Show more

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Highlights

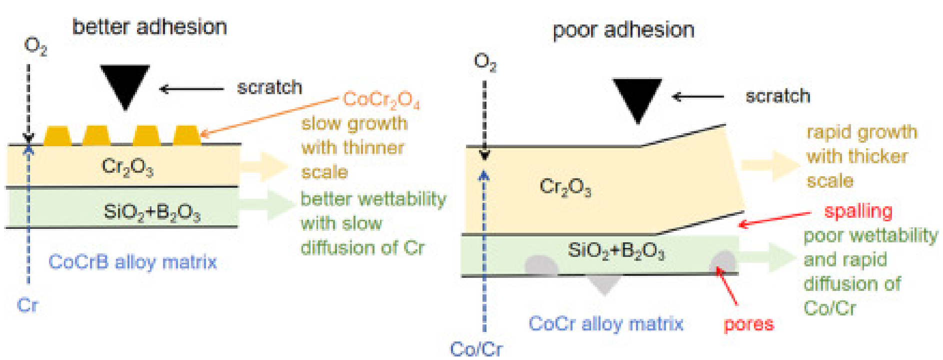
- Pre-oxidation of CoCr alloys with the addition of boron and silicon was carried out to improve the porcelain to metal adhesion.
- The formation of continuous borosilicate layer at scale-alloy interface impeded chromia scale growth with residual spinel nodules.
- Adherence of oxide scale to alloy matrix was enhanced by the addition of B to the alloys.

Outline

Abstract

CoCr-based alloys are widely utilized in dental implants and serve as metallic frameworks for porcelain-fused-to-metal (PFM) restorations, owing to their cost-effective, excellent wear and corrosion. However, interfacial failure between the base alloy and the porcelain arose over extended periods of use in oral environments. Therefore, establishing a reliable metal-porcelain interface is of critically importance. This study investigates how the addition of boron impacts the pre-oxidation scale and its mechanical properties of Co-Cr-W-Cu-Si-(0, 0.5, 1.1 wt.%) B alloys at 928 °C for 10 mins in air. Surface analysis indicated that Si and B stabilized Cr_2O_3 scale grown on the surface of CoCr based alloys. As the B concentration increased, the scale became thinner, and its morphology transformed from Cr_2O_3 to a combination of outer $CoCr_2O_4$ spinel and inner Cr_2O_3. Amorphous SiO_2 was detected along the scale-alloy interface together with B_2O_3. The mechanical stability of the scale along with its adhesion to the matrix increases with increasing B concentration. The synergistic impact of Si and B is probably due to the amorphous innerlayer playing as an effective barrier to the transportation of reactants and slow down the oxide scale growth. Meanwhile, the morphology of pining-like arrangement for amorphous SiO_2 layer increases the factual contact area between the oxide scale and matrix, rendering it less susceptible to fracture.

Graphical abstract



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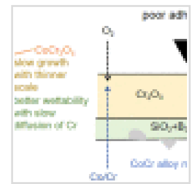

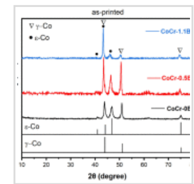

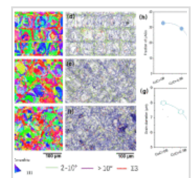

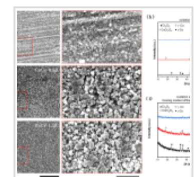

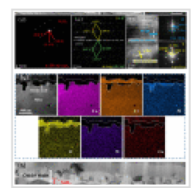

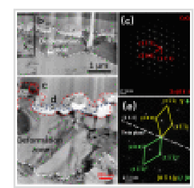

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Tables (4)

Keywords

Porcelain-fused-to-metal; CoCr based alloys; Pre-oxidation; Borosilicate scale

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1. Introduction

Porcelain-fused-to-metal (PFM) restorations, are the combination of metallic skeleton and ceramic veneers, have been extensively used in dentistry, by virtue of their advanced mechanical properties, satisfied clinical results, and widely accepted aesthetics [1,2]. Meanwhile, the cost of metallic scaffold was lowered due to evolution from noble metals to base metals over the time, which makes the PFM more popular, especially in less developed countries. The Co-Cr and Ni-Cr based metals, being composed by non-precious metallic contents, are widely applied in PFM dental restorations. In particular, CoCr alloys become the most appropriate contenders for metallic framework, due to better biocompatibility, excellent wear and corrosion resistance [[3], [4], [5]].

Selected leaser melting (SLM), or 3D printing enables the direct manufacturing of functional parts possessing complex shapes based on three-dimensional digital models. In the dental field, compared with traditional casting techniques, the 3D printing endows implants the freedom of personification and customization, without the trade-off precision [[6], [7], [8]]. Another vital aspect of dental implants is the degree of metal release which relies on manufacturing processes. For CoCrMo alloy, specimen fabricated with suitable SLM parameters show low susceptibility of corrosion and metal release [9]. This is because, although the compositions of as-SLMed and as-cast Co-Cr materials are similar, their metallurgical structures are distinct. The microstructure of as-cast Co-Cr alloy consists of large dendrites, while the as-SLMed Co-Cr alloy holds fine cellular dendrites due to quick cooling and high-temperature gradients [10].

For the successful application of 3D printing CoCr PFM dental prosthesis, the reliable metal-porcelain interface is crucial. In PFM manufacturing process, the ceramic was fused onto the metal framework at elevated temperature (above 900 °C), so an oxide layer will be generated on the surface of matrix framework. During the firing process, the melting porcelain (SiO_2/Al_2O_3/K_2O/TiO_2 etc.) dissolves the outermost layer of the originally in-situ formed oxide layer, with its adhesion to matrix unchanged because inwards and outwards diffusion of Co/Cr/Si/Al/K/Ti from both sides are predominant, especially at scale-alloy interface. Therefore, the quality of originally formed scale, especially the quality of scale-alloy interface, is crucial for the growth of a reliable metal-ceramic bonding [11,12]. For the past decades, the disadvantage of this naturally formed hard-to-control oxide scale, in relation to low bond strength between porcelain and metal have been well-acknowledged [13,14]. Thus, the challenge of promoting the bonding strength between ceramic and CoCr alloys, particularly in cases where oxide scale is inevitably present, persists.

To get a desired bond strength, lots of studies have been focused on various surface treatments to enhance the wetting ability of porcelain on alloys. Among the methods used, pre-oxidation is the straightforward and least costly to alter implant surface [15,16]. The pre-treatment is usually carried out at temperatures up to 1000 °C. However, the role of such pre-oxidation scale remains controversial. Some claimed that it reduced the bond strength between the alloy and porcelain [17]; some reported pre-oxidation didn't impact the metal-ceramic bond strength, but the formation of an excessive oxide layer caused potential distortion of the metal framework; while others thought it enhanced the adherence of ceramic to metal [18,19]. These researchers [20,21] believed that the porcelain powders (SiO_2, Al_2O_3, K_2O) dissolve the outer layer of the grown oxide (normally Cr_2O_3) which is directly bonded to the metal. Such an adherent-to-metal oxide layer becomes the transition zone between the porcelain and metal framework (just like welding).

Consequently, whether the pre-oxidation treatment enhances the bond strength, and how to build a reliable pre-oxidation scale, still needs to be cautiously studied. Essentially, characterizing the pre-oxidized scale is a prerequisite to optimize its performance.

Excessively increasing thickness leads to weakened metal-ceramic bond strength, due to thermal expansion coefficient differences and residual stress. Short pre-oxidation time leads to too thin and discontinuous oxide scale, which is unable to provide sufficient metal-ceramic wetting and chemical exchange. In addition to scale thickness, bonding strength of the scale-alloy interface is also determined by the microstructure of oxide scale. Besides, scale morphology is also vital important for porcelain wetting and mechanical interlocking.

Afterall, the stronger bond strength between oxide scale and alloy, the better for the consequent ceramic-metal bond strength. However, previous research about the high-temperature oxidation of Co-Cr alloys mainly focused on long-term high-temperature treatments, resulting in excessively thick scale. The high-temperature oxidation at a short time scale (10 mins or less, early-stage oxidation) is also essential and must be investigated to achieve further improvement of CoCr-based dental alloys.

Minor alloying element manipulation is an effective way to modify the oxide scale growing behavior and its adherence to the alloy [22]. For the purpose of enhancing the high temperature oxidation resistance of CoCr alloys, several alloying elements were attempted. Pagnano et al. reported that the pre-oxidation was proved to be advantageous merely for when its Mo content exceeded 7 % [23]. The addition of a small quantity of Si has long been known to be beneficial in forming a uniform and dense Cr_2O_3 scale after high temperature oxidation [22,[24], [25], [26], [27], [28]]. It was also discovered that the thickness of oxide scale was reduced as increasing Si content, which in turn, not only releases the potential distortion of the metal framework, but also raising the adhesion between oxide scale and base alloy. Meanwhile, the addition of boron could stabilize the surface chromia scale for CoCr alloys under high temperature conditions. Boron accumulates in the inner oxide scale, significantly enhancing the oxidation behaviour and the adhesion of oxide scale [32]. However, excessive doping could induce the alloy-oxide interface extremely prone to cracking and spallation. Based on what we have known, the impacts of B on the performances of the surface oxide scale formed on CoCr-based dental materials remain unexplained. The synergistic effects of B and Si in affecting scale adhesion and scratch performance have not been studied.

In addition to the benefits of decreased oxide thickness and the improved scale adhesion after alloying with silicon, CoCrWCuSi alloys with excellent antibacterial and mechanical properties were doped with boron and produced by SLM for dental applications. The aim of this research was to further investigate the effects of boron doping, on physical and mechanical properties of the pre-oxidized scale formed on the 3D printed CoCrWCuSiB alloys at simulated sintering condition (980 °C in air for 10 mins). The effect of B doping on the high-temperature oxidation behaviour and scale mechanical properties were evaluated. It is anticipated that the results of this study will assist manufacturers in producing a more reliable and sustainable PFM.

2. Materials and experiments

2.1. Materials

The optimized SLM processing strategy was employed (details can be found in [2]) by mixing commercially customized CoCrWCuSi powder and B powder to fabricate three types of CoCr based alloys, containing 0, 0.5 and 1.5 B (nominal composition, all in wt. %), respectively. The commercially customized CoCrWCuSi powder was purchased from Fujian Zhongke Kang Titanium Material Technology Ltd., China, with average size of 25 μm (5~40 μm). While amorphous nano-sized B powder (with a purity higher than 99.9 %) was purchased from Aladdin Biochemistry Technology Ltd.

The compositions of SLMed alloys were measured by Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES), and the details are listed in Table 1. Hereinafter CoCr based alloys are referred to as CoCr-0B,CoCr-0.5B and CoCr-1.1B, for convenience. Meanwhile, as-printed SLM alloys underwent annealing treatment at 1150 °C for 1 h in a flowing 99.99 % Ar gas. Afterwards, they were gradually cooled inside the furnace for residual stress relaxation. The phase constitution of all alloys was analyzed by XRD.

Table 1. Chemical composition of CoCr-(0, 0.5, 1.1) B alloys.

	Co	Cr	W	Cu	Si	B
CoCr-0B	Bal.	28	9	3	1.5	0
CoCr-0.5B	Bal.	28	9	3	1.5	0.5
CoCr-1.1B	Bal.	28	9	3	1.5	1.1

2.2. High temperature oxidation experiment

As-annealed SLMed specimens in rectangular shape with approximate dimensions of 12 × 6 × 2 mm^3 were used for isothermal oxidation experiments, which were executed in air at 980 °C for 10 mins. The selected temperature mimic with sintering paramters for the dental porcelain. Before oxidation, the surface of each coupon specimen was ground to a 2000 grit finish in water by silicon-carbide abrasive paper, and subjected to ultrasonic cleaning in ethanol before reaction. The hot zone of furnace was calibrated by Omega Calibrator Thermometer with K-type thermocouple. The furnace was heated to 980 °C and stabilized for 20 mins, and the specimens were then pushed into the center of the validated hot zone. After 10 mins, the furnace was shut down and the specimens were allowed to cool in the air overnigh until room temperature was reached.

2.3. X-ray photoelectron spectroscopy (XPS) analysis

The chemical state for elements of interest and their content were measured using X-ray photoelectron spectroscopy (XPS, ESCALAB 250Xi-Thermo Scientific) for the oxidized specimens. The depth profile was obtained by sputtering with Ar ion and the sputtered region was 2 × 2 mm^2, while the rate of sputtering was 0.66 nm/s for a standard Ta_2O_5 sample. The XPS analysis was performed on an circular area of 0.5 mm in diameter. The etching and spectrum acquisition procedures were performed in an alternating manner to acquire the depth profiles.

2.4. Scratch testing

Scratch testing was carried out after the high temperature oxidation experiments, by employing the Anton Paar STEP500 Rockwell indenter with a 100 μm radius tip at ambient conditions, and without lubricant between the specimen and the indenter. The micro-scratch tester is also fitted with an acoustic emission (AE) which allows the quantification of the critical force at which the first observable crack in the scale appears. The normal force being applied was raised from 30 mN to 30 N while the sliding speed was 8 mm/min and the sliding distance was 4 mm. The indenter experienced solely vertical displacement, while the specimen holder was moved horizontally. Meanwhile, the scratch tests were carried out in triplicate to evaluate reproducibility of the results. After performing statistical analyses of these forces, it was found that there is a positive correlation between acoustic emission and optical microscopical evaluation.

2.5. Microstructural characterization

Specimens were ground with silicon carbide papers down to #2000 grit, followed by polishing with 3, 1, and 0.05 μm diamond suspensions, stepwisely. Scanning Electron Microscopy (SEM, FEI Quanta 250), Focus Ion Beam (FIB, G5 UX-Thermo Scientific), Transmission Electron Microscopy (TEM, Philips CM200), X-ray Energy-Dispersive Spectroscopy (EDS, Bruker), and Electron Backscatter Diffraction (EBSD, FEI-XL50) were carried out for the analysis of microstructures. For the EBSD analysis, the specimen was tilted at an angle of 70 °C, while maintaining an operational distance of 17 mm with different step sizes.

The phases of the specimens were characterized by X-Ray Diffraction (XRD, PANalytical Empyrean) with Cu Kα radiation. Particularly, Grazing Incidence X-Ray Diffraction was used to characterize the crystalline structures of the oxide scales that developed on the surfaces of the CoCr alloys. The angle of incidence was started at 2°, and the diffraction patterns were documented for 2θ values in the range of 10° to 80°.

3. Results

3.1. The microstructure of as-printed CoCr alloys

The XRD patterns of as-printed alloys were depicted in Fig. 1. Both diffraction peaks of FCC γ-Co phase and HCP ε-Co phase can be identified for all these alloys. The boundaries of the melt pool and the scan track are clearly observed from the X-Y plane EBSD morphology of as-printed CoCr samples (Fig. 2). At the macro-level, the perpendicular cross-section showed a compact homogeneous cellular microstructure. As the laser was scanned backwards and forwards in tracks, melting happened successively. When manufacturing parts in a layer-by-layer manner, each line of material experiences gradual melting and recrystallization. Grains re-melts and re-grows in the overlapping zone. Consequently, it can be anticipated that the resultant structure was fine-grained and resemble the structure described in the literature [33]. No obvious preferred crystallographic orientation effect was observed for all as-printed alloys. The X-Y plane grain size distribution of all alloys was also derived from EBSD results and is depicted in Fig.2(g). It is obvious that the grains of CoCr-1.1B are slightly smaller than those of other specimens. The average grain diameter of 0B, 0.5B, and 1.1B was 8.0 μm, 7.4 μm, and 5.7 μm, respectively. It is evident from Fig. 2(h) that the fraction of high angle boundary is reduce with increasing the B content. This indicates that the B can refine the microstructure.

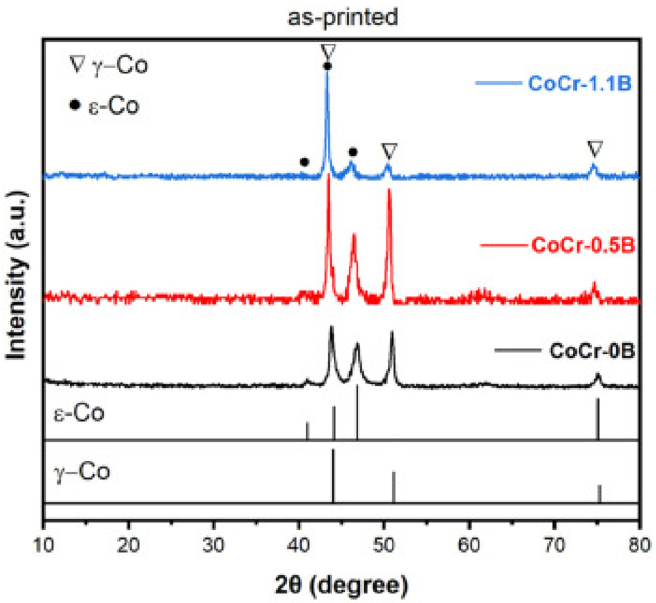

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Fig. 1. The XRD pattern of as-printed alloys.

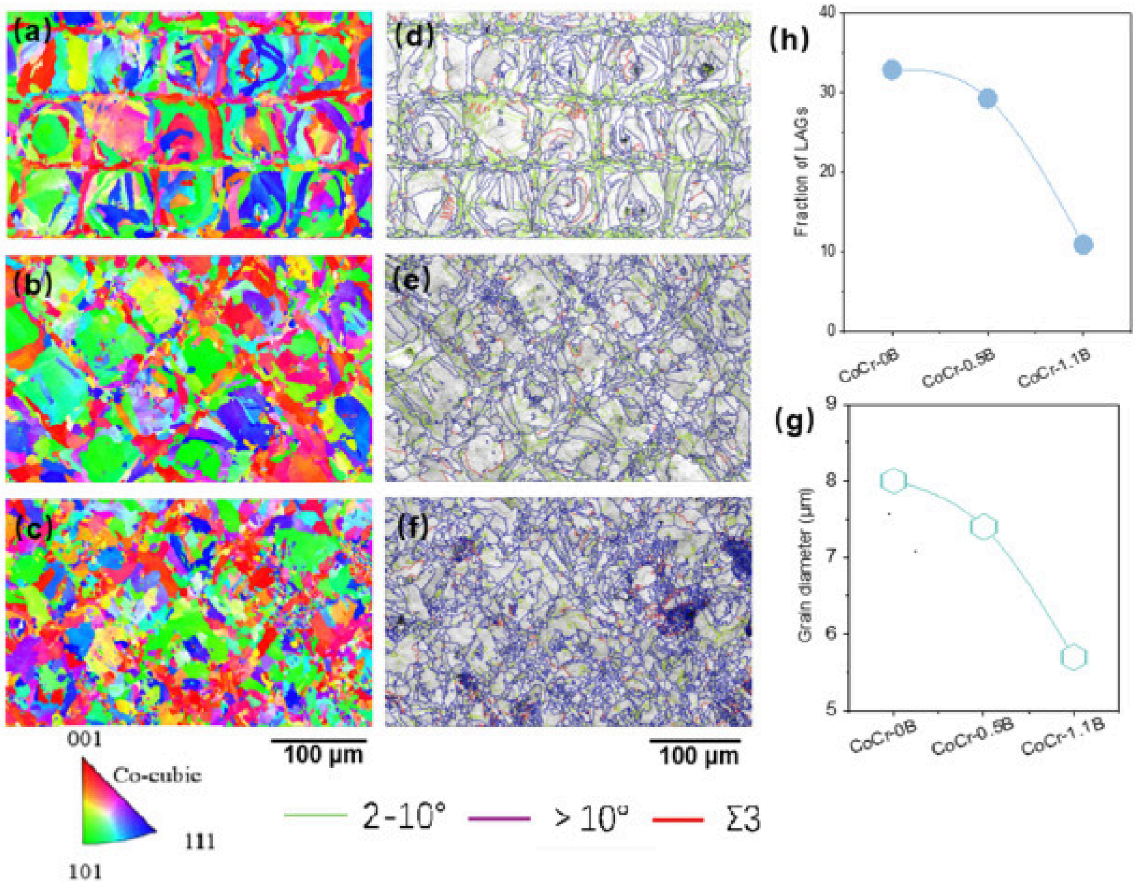

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Fig. 2. EBSD Inverse Pole Figure (IPF) maps of X-Y plane for (a) CoCr, (b) CoCr-0.5B, (c) CoCr-1.1B; grain boundary maps for (d) CoCr, (e) CoCr-0.5B, (f) CoCr-1.1B; (g) the average grain diameter (triangle colour codes referred to color in this figure legend); (h) the fraction of high angle boundaries.

3.2. Scale morphology

The CoCr-(0, 0.5, 1.1) B alloys were subjected to oxidation in air at 980 °C for 10 min and the SEM surface micrographs are illustrated in Fig. 3(a–f). The corresponding XRD patterns in Fig. 3(Fig. 3, Fig. 3) help to identify the phase structure of the reaction products. From top surface, it is seen that the chromia scale on the B-free CoCr alloy is predominantly consisted of fine oxide grains, whereas the scales on the CoCr-0.5B and CoCr-1.1B alloys displayed additional flake-shaped platelets on the surface. The XRD analysis (Fig. 3(Fig. 3, Fig. 3)) confirmed chromia as the dominant phase for all three alloys. Additionally, grazing incidence X-ray diffraction (Fig. 3(i)) revealed the existence of $CoCr_2O_4$ spinel on the CoCr-0.5B and CoCr-1.1B alloys. The XRD profiles also indicated that the matrix of all three alloys is consisted of γ-Co and ε-Co phases. However, as XRD primarily detects the uppermost few micrometers of the surface, particularly in grazing incidence mode, it predominantly reflects surface characteristics. By comparison with the bulk XRD analysis in Fig. 1, it is evident that γ-Co dominates over ε-Co in the subsurface layer, corresponding to the delimination of the deformation induced ε-Co due to recrystallization at high temperature. The adhesion between CoCr alloys and substrate has an inverse proportion to the hardness of the interfacial layer. The dual-phase coexistence of γ-Co and ε-Co provides strength to the metal framework, while single-phase γ-Co contributes ductility. Such a balance enhances metal-ceramic bonding by relieving stress and promoting better adhesion.

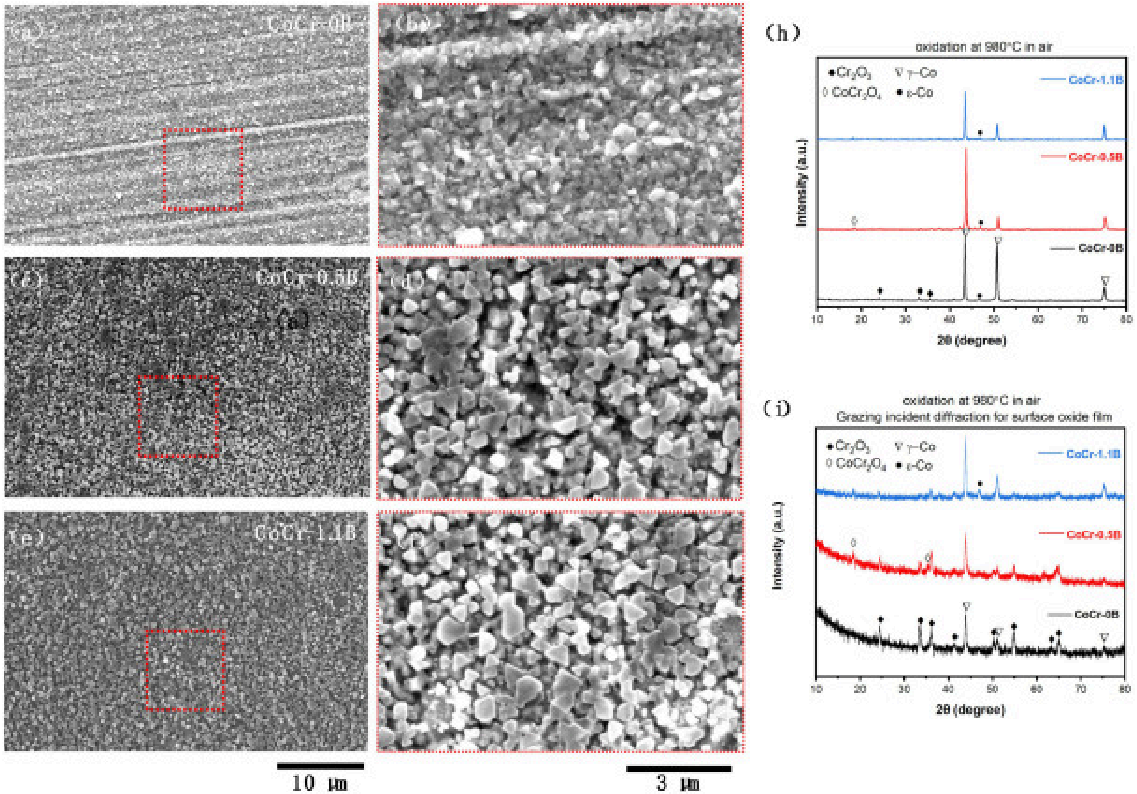
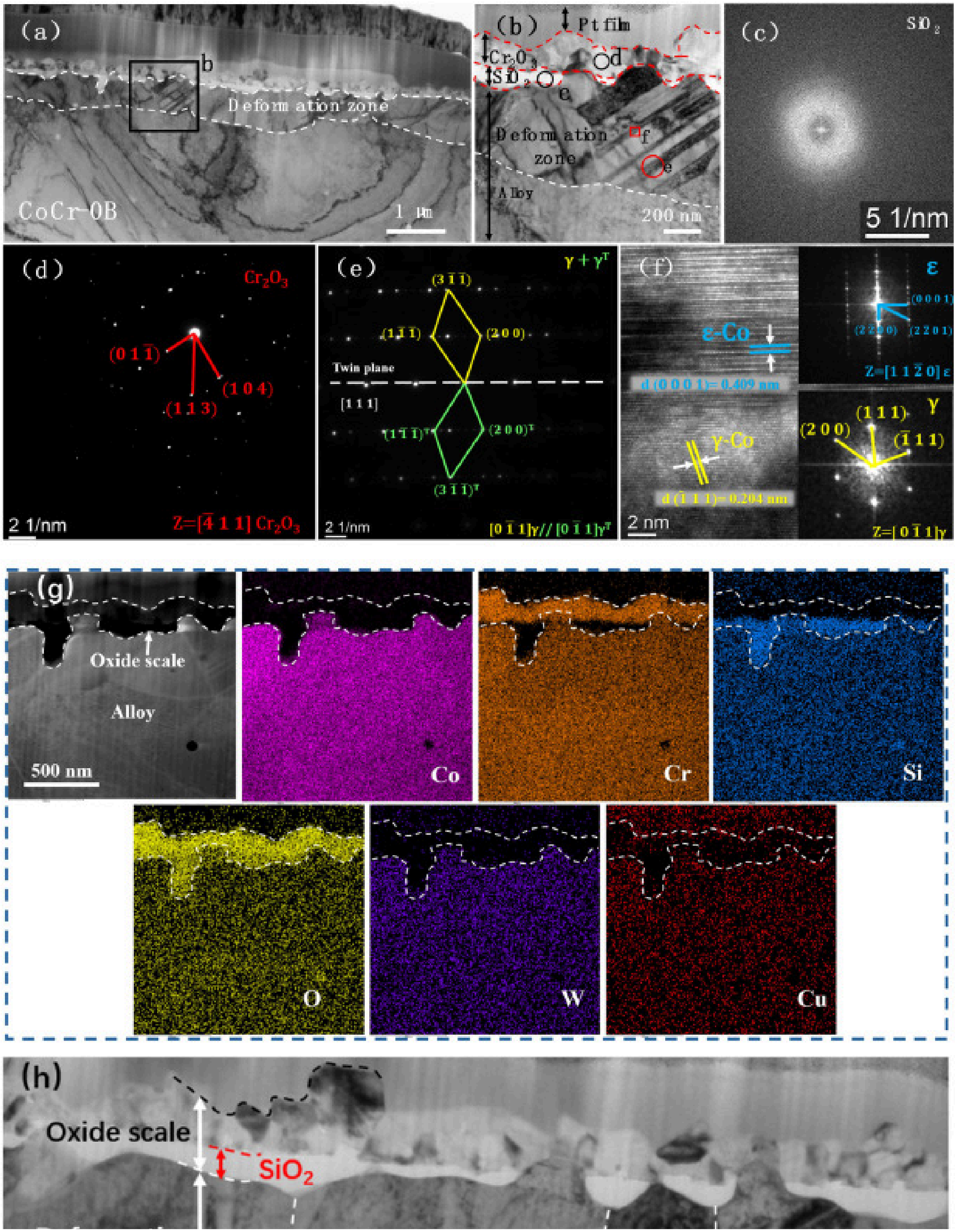


Fig. 3. The SEM top view morphologies of (a-b) CoCr, (c-d) CoCr-0.5B, (e-f) CoCr-1.1B and (h, i) the XRD patterns after high temperature oxidation in air at 980 °C for 10 mins.

The bright field (BF) TEM cross-sectional images of the oxide scale formed on B-free CoCr alloy reacted at 980 °C for 10 mins in air are shown in Fig. 4. The enlarged details and corresponding SAD pattern (Fig. 4(d)) together with energy dispersive X-ray spectroscopy (EDS) elemental mapping analysis (Fig. 4(g)) confirmed that the outer Cr_2O_3 scale is dominant with fine equiaxed grains, and a thin discontinuous SiO_2 scale was observed at the interface with EDS evidence (Fig. 4(g)). At some specific locations, SiO_2 is also observed to develop inwards along the grain boundaries in the underneath deformation zone (Fig. 4(h)). A deformation zone about 500 nm in depth underlying the oxide scale is visible (Fig. 4(d)), with smaller grain size compared with the grains deep in the matrix. Notably, grains dominated by γ-Co deformation twins with scattered fine ε-Co laths were found within the deformation zone. These two sutble features can be identified by high resolution image (Fig. 4(b, e, f)). The subsurface deformation zone should be introduced by abrading the specimen on silicon sand paper, and recrystallisation occurred due to oxidation carried out at high temperature. Deformation-recrystallisation accounts for the refined structure over this zone. The equilibrium phase diagram of CoCr alloy demonstrates that the predicted phase structure is FCC γ-Co above 900 °C and HCP ε-Co below 900 °C. In general, the FCC structure exhibits ductility, and the HCP leads to more brittleness. In this work, γ-Co and ε-Co were equally present deep in the alloy matrix during cooling, while the deformation induced γ-Co twins and competitively reduced the presence of ε-Co to some extent, providing better ductility for metal-ceramic interfacial bonding.


(a)
b
Deformation zone
CoCr-0B
1 μm
(b)
Pt film
Cr_2O_3
SiO_2
d
c
Deformation zone
f
e
Alloy
200 nm
(c)
SiO_2
5 1/nm
(d)
Cr_2O_3
(0 1 $\bar{1}$)
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(1 1 3)
2 1/nm
Z=[$\bar{4}$ 1 1] Cr_2O_3
(e)
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(3 $\bar{1}$ $\bar{1}$)
(1 $\bar{1}$ $\bar{1}$)
(2 0 0)
Twin plane
[1 1 1]
(1 $\bar{1}$ $\bar{1}$)^T
(2 0 0)^T
(3 $\bar{1}$ $\bar{1}$)^T
2 1/nm
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(f)
ε-Co
d (0 0 0 1)= 0.409 nm
γ-Co
d ($\bar{1}$ 1 1)= 0.204 nm
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ε
(0 0 0 1)
(2 2 0 0)
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Z=[1 1 $\bar{2}$ 0] ε
γ
(1 1 1)
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Z=[0 $\bar{1}$ 1]γ
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Oxide scale
Alloy
500 nm
Co
Cr
Si
O
W
Cu
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Oxide scale
SiO_2

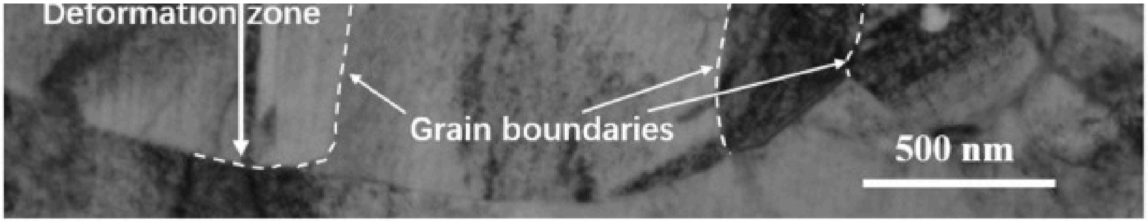

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Fig. 4. TEM bright field (BF) cross-sectional images of B-free CoCr based alloy after reaction in air at 980 °C for 10 mins (a) and (b), the corresponding selected area electron diffraction (SAD) pattern showing amorphous SiO_2 (c), Cr_2O_3(d), γ-Co twin morphology (e), high-resolution image of ε-Co (f), EDS elemental mapping of oxide scale (g) and enlarged oxide scale showing SiO_2 partially elongated at substrate grain boundaries (h).

Fig. 5 shows BF TEM cross-sectional images of a thin scale formed on CoCr-0.5B alloy reacted in air at 980 °C for 10 mins, as well as the corresponding SAD patterns and high-resolution image with FFT pattern. The scale is consisted of three distinctive layers. There is an discontinuous outer layer of coarse-grained $CoCr_2O_4$ spinel which is correlated with the flake-shaped plates on the top surface (Fig. 3). An intermediate layer of fine-grained Cr_2O_3 (Fig. 5(b)) is found to be continuous, and inner amorphous SiO_2 layer is inconsecutive (Fig. 5(b)). A subsurface deformation zone is found about 1 μm deep underneath the scale, in which ε-Co grains are sparsely distributed among γ-Co with deformation twins to some extent (Fig. 5).

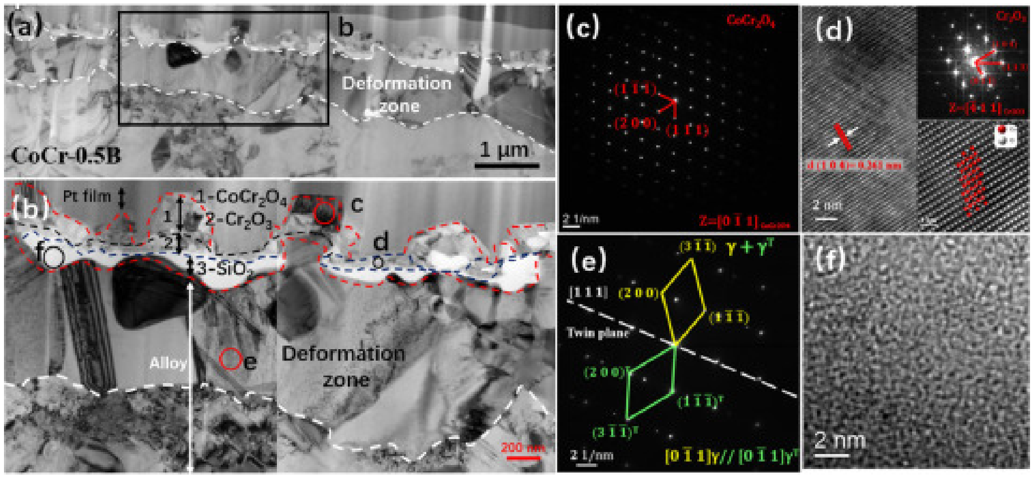

Fig. 5. TEM bright field (BF) cross-sectional images of CoCr-0.5B after reaction in air at 980 °C for 10 mins (a) and (b), the corresponding SAD/high-resolution image showing (c) $CoCr_2O_4$, (d) Cr_2O_3, (e) γ-Co deformation twins, and (f) amorphous SiO_2.

The BF TEM cross-sectional images of the thin layer formed in air at 980 °C on CoCr-1.1B alloy are exhibited in Fig. 6. Similar to the layer formed on the CoCr-0.5B alloy, there are more densely distributed $CoCr_2O_4$ plates on the top of fine-grained chromia layer, followed by the discontinuous glassy SiO_2 underneath the chromia. The subsurface deformation zone was also detected beneath the scale, showing a bit of ε-Co in addition to γ-Co deformation twins.

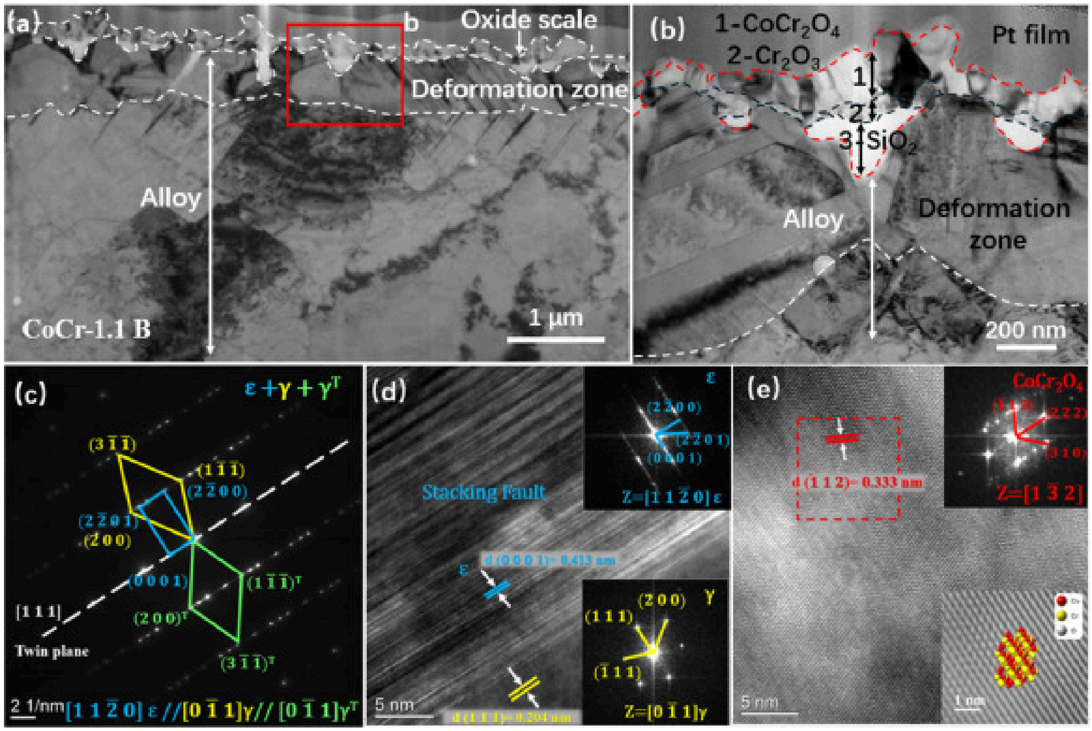

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Fig. 6. TEM micrography showing cross-sectional BF image (a, b), SADP (c) and high resolution image (d, e) of CoCr-1.1B reacted for 10 mins in air at 980 °C.

The thickness of the oxide scale was statistically calculated according to the BF TEM cross-sectional images, and the thickness values are presented in Table 2. The average thickness of the oxide scale shows a gradual reduction with the addition of B. It is 0.23µ m for the B-free CoCr alloy, 0.21µ m for CoCr0.5B alloy, and only 0.18µ m for CoCr1.1B alloy. Due to the significant fluctuation of scale morphology, especially considering the existence of $CoCr_2O_4$ spinel nodules at early stage of oxidation, thickness data deviation is relatively large; however, the trend is statistically evident and meaningful to understand the short-term scale formation. Unlike the excessive oxidation over long-term, the scale is basically thin but its growth rate can be high at the initial stage, while internal stress accumulates at the scale-alloy interface due to the fast-growing oxide scale. The reduction of scale thickness, associated with slower growth rate, by the addition of B, may relieve the internal stress and improve the adhesion between scale and alloy matrix. This suggests that the addition of boron at a low level promotes the high-temperature oxidation resistance of the CoCr-based alloys.

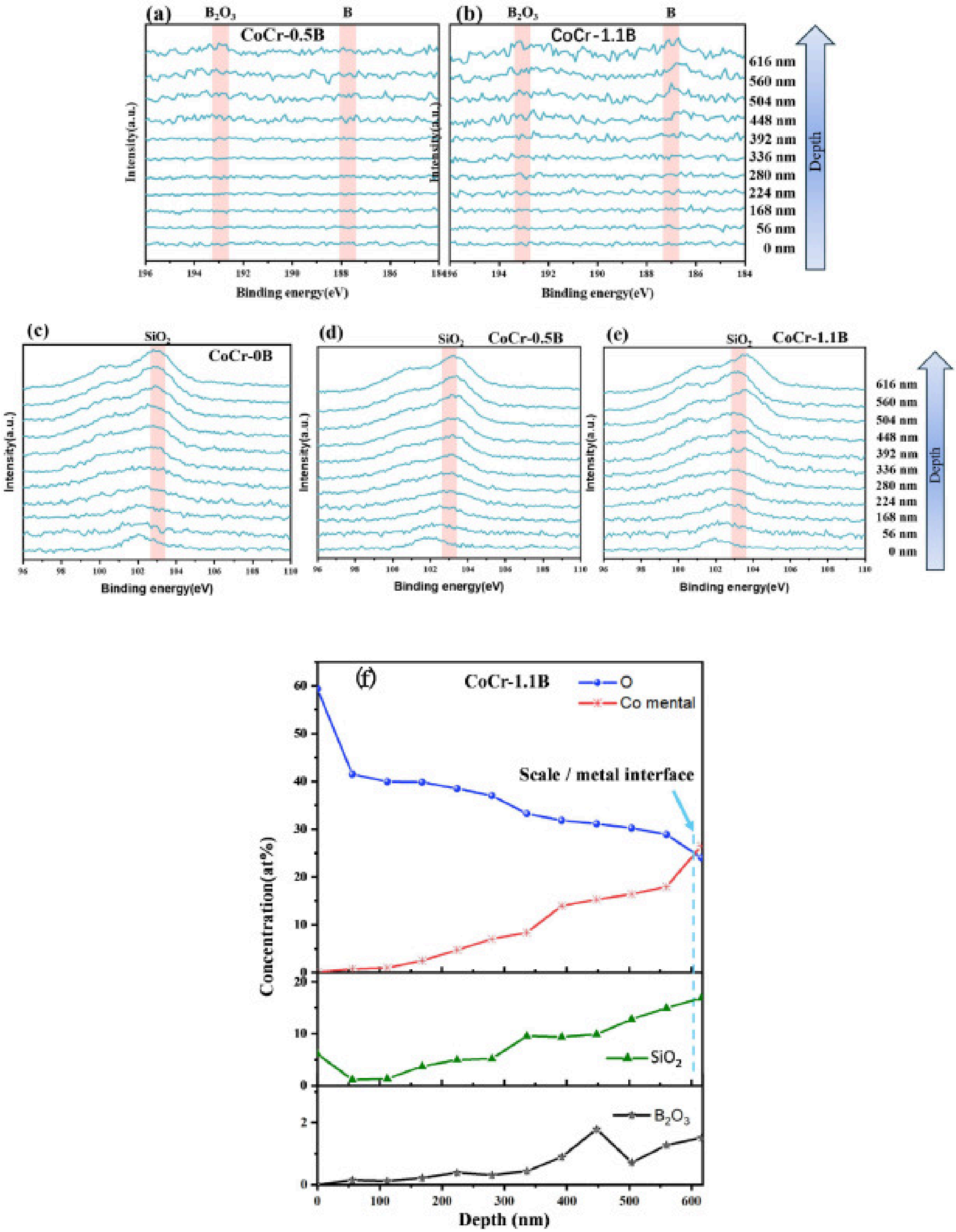
Table 2. Average oxide scale thickness for CoCr-(0, 0.5, 1.1) B alloys.

Alloy	Thickness (μm)	Standard deviation (μm)
CoCr-0B	0.23	0.047
CoCr-0.5B	0.21	0.059
CoCr-1.1B	0.18	0.091

3.3. XPS depth profile analysis

In this study, XPS has been utilized for all alloys reacted in air for a short duration of time (10 mins) to investigate the surface compositions and depth profile for the elements of concern. For the light elements, like boron, the quantitative analysis are not straightforward, and problem may also arise when applying EDS in SEM and TEM. It is seen from the XPS spectra in Fig. 7 that the B1s binding energy (BE) of around 193 eV can be assigned to B_2O_3 [34], whereas the additional peak appear at around 188 eV is reported for boron atoms dissolved in the metal matrix [35]. The formation of B_2O_3 layer is evident at a depth of around 600 nm for B-bearing CoCr alloys and particularly for the higher boron content alloy (CoCr1.1B), as presented in Fig. 7(b). The point of intersection between the depth profile of O 1 s and metallic Co was suggested corresponding to the scale-alloy interface. Based on XPS depth profiling analysis (Fig. 7(f)), the scale-alloy interface locates at around a depth of 600 nm for CoCr-1.1B alloy. The concentration depth profile of Si reaches maximum at around 500–600 nm (Fig. 7(d–f)). It is noted that the depth profile was calculated based on the sputtering rate of 0.66 nm/s for a standard

Ta_2O_5 sample, and the ture scale thickness can be read by microstructural analysis in Fig. 2. The BE (Si 2p) of around 103 eV can be assigned to SiO_2. Accoding to literature[36], the 103 eV was for Si_2O_3, possibly due to the diffusion and dissolution of Si into Cr_2O_3 phase outwards from the interface. It indicates that the boron is segregated within the inner layer of the oxide scale, and closely next to the scale-alloy interface, which may overlap with the SiO_2 layer to some extent. Furthermore, atomic percentage of B_2O_3 at the interface can be as high as 1.5 % based on XPS survey, far higher than it appears at outer layer of the oxide scale. The formation of an inner layer of SiO_2 with B-doping is likely to delay the inward diffusion of Cr with lower growth rate, accordingly [37].


(a)
B_2O_3
B
CoCr-0.5B
(b)
B_2O_3
B
CoCr-1.1B
616 nm
560 nm
504 nm
448 nm
392 nm
336 nm
280 nm
224 nm
168 nm
56 nm
0 nm
Depth
Intensity(a.u.)
Binding energy(eV)
(c)
SiO_2
CoCr-0B
(d)
SiO_2
CoCr-0.5B
(e)
SiO_2
CoCr-1.1B
(f)
CoCr-1.1B
O
Co mental
Scale / metal interface
SiO_2
B_2O_3
Concentration(at%)
Depth (nm)

Download: Download high-res image (1MB)

Download: Download full-size image

Fig. 7. X-ray Photoelectron Spectra in the B1s region acquired at different depths for (a) CoCr-0.5B, (b) CoCr-1.1B; and Si1s region acquired at different depths for (c) CoCr-0 B, (d) CoCr-0.5B; (e) CoCr-1.1B, respectively, and (f) XPS depth profiles for CoCr-1.1B.

3.4. Adhesion of oxide scale

Optical images of the residual cracks and acoustic emission with normal load F_n of all specimens in this work are shown in Fig. 8. The orientation of scratch is indicated by the arrow. When subjected to small loads at the beginning of scratching operation, acoustic emission remains low with a flat line, and the scale undergoes elastic-plastic deformation without rupture (Fig. 8(d)). Afterwards, the acoustic emission increases sharply as the normal load increases and fluctuates after this critical load. The first sharp rise of acoustic emission in the critical loads in Fig. 8 reached 7.79 N, 10.49 N, and 7.94 N, for CoCr-0B, CoCr-0.5B, and CoCr-1.1B, respectively. Statistical evaluations were carried out on these critical forces with the aim of excluding outliers and revealing critical loads of significance. A significant correlation was discovered between acoustic emission and optical microscopical analysis. The critical loads would induce scale spallation and expose the metal in particular regions. Correspondingly, as a consequence of stress whitening, the bright area was produced in the centre of scratch grooves [38]. At supercritical conditions, spalling/chipping/parallel and transverse cracks can be observed (Fig. 8(a)).

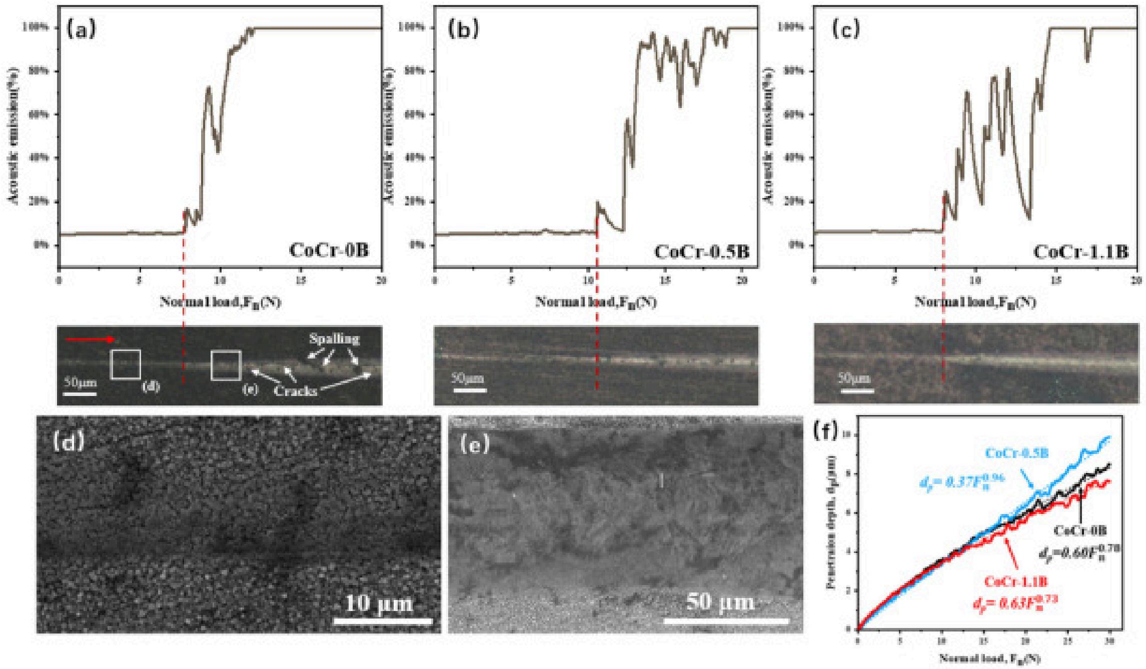


Fig. 8. Acoustic emission and optical images of residual scratch morphologies of (a) CoCr-0B, (b) CoCr-0.5B, (c) CoCr-1.1B, (d-e) enlarged detail of residual scratch in (a), and (f) variation of penetration depth d_p with normal load F_n.

The variation of penetration depth d_p with normal load F_n are depicted in Fig. 8(f). As can be observed, the d_p increases non-linearly with F_n, and the relationship between d_p and F_n are adaptable to the power-law functions [39]. The determined critical values are presented in Table 3. The scale adhesion results derived from scratch tests are provided as normalized critical pressures $p_{CritNorm}$, calculated using the method illustrated in literature [40]. Based on the certain critical loads, the pressure exerted in N/mm² was computed:

$$p_{Crit} = \frac{F_n}{A_{sc}} = \frac{F_n}{\pi \cdot r \cdot d_p} \left[Nmm^{-2}\right]$$

where A_{sc} represents the contact area [mm²], r corresponds to the radius of the Rockwell-C sphere [mm], d_p is the penetration depth [mm], and F_n is the critical failure force [N]. As the scale thickness differs among specimens, the critical pressure was normalized to the oxide scale thickness t [mm] (Table 2) which is calculated from

$$p_{CritNorm} = \frac{F_n}{\pi \cdot t \cdot r \cdot d_p} \left[Nmm^{-3}\right]$$

Table 3. Parameters used to calculate scale bonding strength.

	CoCr-0B	CoCr-0.5B	CoCr-1.1B
F_n **The critical failure force in N**	7.79	10.49	7.94
d_p **The indented depth in mm**	2.96	3.55	2.89
P_{Crit} [Nmm^{-2}]	8376.17	9409.29	8761.15
$P_{CritNorm}$ [Nmm^{-3}]	34,975.03	43,134.19	46,138.00

Based on the normalized critical pressures values $p_{CritNorm}$ in Table 3, the adhesion and failure at the boundary of the oxide scale and matrix could be measured quantitatively, and statistically analyzed for varying boron alloying concentration, demonstrating the mechanical stability of oxide scales formed by pre-oxidation. The influence of B-doping on the adhesion of oxide scale can be deduced, that boron has a tendency to slightly enhance the adhesion. With increasing B content in the alloy, the stability of oxide scale become higher.

4. Discussion

Pre-oxidation of B-bearing CoCrWCuSi alloys under air-based conditions resulted in diverse scale morphologies according to the boron concentration in the alloy. The oxidation products observed in this study are summarized in Table 4. The oxidation behavior of CoCr alloys including 0, 0.5, and 1.1 wt % B, as well as the effect of trace alloying elements on the physical and mechanical properties of the oxide scale, are now disccussed.

Table 4. Pre-oxidation products of the studied alloys.

	CoCr-0B	CoCr-0.5B	CoCr-1.1B
Outer layer	/	Co-Cr spinel	Co-Cr spinel
Intermediate layer	chromia	chromia	chromia
Inner layer	Discontinuous silica	Discontinuous silica + boron oxide	Discontinuous silica + boron oxide

4.1. The role of subsurface deformation layer

Prior to high temperature oxidation experiment, specimens were prepared through standard metallographic procedures, including cutting and abrading. During this step, additional dislocations accompanied by residual stress were induced in the form of a cold worked subsurface region, especially causing stress concentration at grain boundaries due to dislocation pile-up. Even under elevated temperature conditions, the stacking fault energy (SFE) of CoCr based alloy system remains significantly low [41,42]. The presence of austenitic twins could be induced by deformation of the subsurface. However, the effect of cold work on the specimen was not uniform, and consequently the deformation twin facilitated by dislocations and stress varied across the surface. The deformation zone with small grains containing ε-Co together with austenitic deformation twins (Fig. 4, Fig. 5, Fig. 6) was observed beneath the thin oxide scale. The oxidation of chromia-forming alloy is partially determined by the chromium supply through alloy matrix, especially grain boundary. The deformation zone where diffusion occurs much more rapidly was also shown by earlier workers for the chromia growth during high temperature oxidation [25,43]. Thus, all the three studied CoCr based alloys in this work were able to form a dense and continuous thin chromia scale under short exposure in high temperature air.

It is known that the diffusion coefficient of iron in amorphous SiO_2 is quite small which suggests that the diffusion coefficient of cobalt in SiO_2 is even less due to the similarity of these two metallic elements [44]. In the case of B-free CoCr alloy, the fine-grained deformation zone beneath the thin scale provided the fast Cr supply for the chromia scale growth. Then the early formed $CoCr_2O_4$ spinel could be spalled off, resulting in a quick leap from transient stage to establishment of a stable chromia scaling within 10 mins exposure at 980 °C in air. Whereas in this situation, the deformation zone beneath the scale can be depleted in chromium and may cause not only chromia scale failure in the long run, but also degrade alloy mechanical integrity to some extent [25]. Moreover, the fast growing chromia may lead to excessive growth stress in the scale, raising mechanical damage potential to the scale. It is worthnoted that the addition of B to the alloy does not change the formation deformation zone with fast Cr diffusion, but B-doping slows down the Cr diffusion as well as the scale growth in another and more significant manner, as stated in Section 4.2.

4.2. The effect of B on oxidation

The beneficial effects of small amount of Si addition to CoCr alloys was long been known. The SiO_2 layers serve as efficacious barriers to the transportation of oxygen and metal cations, resulting in the reduced growth rate of the overlaying solid chromia layer [45]. However, disadvantages for Si include detrimental effect of scale spallation especially for austenitic steels [24,46]. Also, some reported the addition of Si reduced the surface roughness of the oxide surface, which potentially caused the adhesion between the metal substrate and porcelain to become poorer due to a decline in the anchoring effect.

In this work, the SiO_2 layer is observed to be discontinues, and elongated especially along the matrix grain boundaries or cleaves between the neighbouring deformed grains. The

SiO_2 layer is visualized as pins penetrating into the metal substrate in a sort of dovetail arrangement (as shown in Fig. 4, Fig. 5, Fig. 6). Because the penetrated portion resembles a nail that tightly binds the oxide scale and the metal together, it is aptly referred to as "pinning" effect [47]. Furthermore, this morphology not only relieves the spallation susceptibility, but also increases the effective contact area between the oxide scale and the metal, thereby could partially contribute to enhancing the adhesion. In fact, the pinning points were originally dented hollows on surface, possibly as a result of abrading and deformation. Non-uniform metal surface can entrap air and contaminants, leading to porosity formation in the porcelain during sintering. This would cause a decrease in the contact area and bond strength. Therefore, it is speculated that glassy silica filling vacancies at the oxide-metal interface may play a fundamental role in the enhancement of the oxide adherence.

On the other hand, for the B-doping CoCr alloys, especially for 1.1 % B doping, the segregation of B in the form of B_2O_3 is more intense at the scale-metal interface, which overlaps part of the amorphous silica layer, as can be seen from XPS depth profile (Fig. 7). Scaling is then influenced by diffusion through a thin silica layer together with the boron oxide at the chromia-alloy interface (Fig. 7), serving as an additional diffusion barrier. In turn, slowly growing chromia layer could lessen the need for high Cr supply capabilities, and reliving the internal growth stress caused by scaling. Therefore, the addition of B to CoCrWCuSi alloys reduced the thin scale growth rate as can be seen from scale thickness statistical evaluations (Table 2). The higher level of B-doping, the thinner of scale developed on the surface after oxidation at 980 °C for 10 mins. Notably, with the boron alloying, the surface oxidation products switched from chromia to the inner chromia with ourter CoCr spinel structure (as listed in Table 4). Glassy silica could fill the micro-cracks of deformation zone, providing better adhesion for the subsequent scale growth. The co-alloying of B and Si may further modify the growth of silica layer via eliminating the micro-cracks. In borosilicate glass production industry, B promotes glass melting with lower viscosity at lower temperature, which is favourable to get rid of bubble for the following shaping process [48,49]. The greater connectivity between borate and silicate sub-networks in the amorphous structure, the better thermal and chemical stabilities demonstrated by the glass. For borosilicate glass, trivalent B is known to be added to the tetravalent-silica sol to produce a non-periodic structure, and this alterable and complicated Si-O-B structure is in a position to decrease the residual stress and thermal mismatch stress, so as to boost the coatings' ductility [50]. Overall, co-alloying of B and Si facilitates the formation of a thin protective chromia layer and its adherence to the substrate [51].

4.3. The formation of $CoCr_2O_4$ spinel

With focus on CoCr-based alloys as heat-resistant materials, their high-temperature oxidation properties have been extensively studied. Researchers examined the oxidation behaviour of Co-10Cr alloy at 1073-1573 K and noticed duplex oxide layers constituting of an outer CoO scale and an inner scale of Cr_2O_3 and $CoCr_2O_4$ being present [45]. Raising up the Cr content brought about the formation of Cr_2O_3 scale. The formation of Cr_2O_3 scale was explained by the selective oxidation of Cr in an environment with low oxygen partial pressure. Nevertheless, CoO was still discovered in the CoCr alloy during the early period of the high-temperature oxidation. Spinel-like $CoCr_2O_4$ is regarded as forming via a reaction between CoO and Cr_2O_3. Large amounts of $CoCr_2O_4$ were detected in the present study (Fig. 5, Fig. 6).

During the transient period, a very initial stage of oxidation, Cr in the CoCr alloys is insufficient, together with the low diffusion coefficient of chromium resulting in occurrence of cobalt oxidation [52]. It is suggested that cobalt exists at higher concentration than Cr in the bulk CoCr alloy, thus cobalt oxides developed on the surface at the very beginning of oxidation. A further consequence was $CoCr_2O_4$ spinel formation with increasing chromium content. With longer duration of oxidation, CoO and $CoCr_2O_4$ are progressively substituted by Cr_2O_3, forming a single-layered oxide scale. Due to the subsurface deformation zone where a sufficient supply of Cr to the surface is sustained, then these precipitates of Cr_2O_3 laterally consolidate to form a continuous scale of Cr_2O_3. Simultaneously, the additional presence of SiO_2 scale reduces the oxygen partial pressure at the matrix surface. This suppresses the formation of cobalt oxide while promoting the formation of chromium oxide. There had also been spinel formation for B-free CoCr alloy at the initial stage; however, as long as it transits to the stage of selective oxidation, spinel could spall off and that is why only Cr_2O_3 scale was eventually observed at the moment of 10 mins (Fig. 4).

4.4. Scale adhesion and implication for PFM bonding

It was found from this work that co-alloying with boron tends to increase the scale adhesion. The mechanical strength of the oxide scale and its adherence to the substrate are given by calculating the normalized critical pressures $p_{CritNorm}$, as listed in Table 3. With more boron doped in the alloy, a thinner oxide scale was produced inthe same oxidation period, which enhances stability of the oxide scale. Also, the resulting oxide scale structure varies with boron concentrations, partially casting influence on the adhesion. Thick chromia layer together with underneath amorphous silica layer is more brittle and spalling off easily compared to a thin sandwich structured layer (spinel on the top with fine-grained chromia in the middle, followed by the mixture of silica and boron oxides). In general, a higher layer density correlates with a better surface adherence and an enhanced scale stability. This is consistent with previous studies [[29], [30], [31], [32]] and it is suggested that alloying of boron to CoCr plays a significant role in the oxide scale formation and its properties. Moreover, the beneficial effects of boron addition could more effective when silicon is present because both of them segregate at the interfaceS. Based on this, it is suggested that the synergistic influence of SiO_2 and B_2O_3 formation on improving oxidation resistance and oxide scale adhesion is attributed to the SiO_2 and B_2O_3 layer, playing as barriers to reactants transport, and slowing down the oxide scale growth. This phenomenon could influence mechanical properties of the scale, making it less susceptible to fracture.

Good adhesion between the oxide scale and metal as a result of pre-oxidation is a prerequisite for good adhesion between porcelain and metal in the following sintering process. Additionally, there is an transitional layer formed between metal and ceramic, determining the bonding strength of porcelain sintering. The reaction between matrix and ceramic significantly influences the transitional layer. The transitional layer becomes thicker as the interaction area broadens. Elements such as Co and Cr migrated from metal matrix to porcelain, while O, Si, Al, and K migrated in the opposite direction [21,53]. Consequently, the transitional layer was inferred to be the location of chemical reaction and elemental exchange.

It is a well-known fact that the bond strength correlates directly with the thickness of the oxide scale. Poor bonding strength results from the absence, extremely thin or thick oxide scale. Wu et al. [14] described that the transitional layer in the SLM CoCr alloy had a thickness of roughly 7 to 10 μm, though the initially formed oxidation scale could be much thinner. However, the formation of growth stresses in the oxide scale could be one of the factors that affect the adherence of the oxide scale. The CoCr-1.1B alloy in this work exhibited oxide scale thinner than 1 μm. Proper thickness could be eventually obtained by the slowly oxidizing rate, which is also associated with smaller stress and better adhesion, compared to the undoped CoCr alloy. Furthermore, discontinuous $CoCr_2O_4$ spinel intrusions left on the top surface of the scale provides an opportunity for mechanical interlocking between the porcelain and the matrix. Therefore, the complicated oxide morphology jointly improve adhesion: spinel shows roughened surface, and chromia has protective oxidation. Furthermore, the adhesion properties are also influenced by the chemical state of an oxide scale, which is altered by the addition of Si/B. This is because the bond between the substrate and porcelain primarily results from the chemical reactions between the metallic oxides and porcelain. Cobalt in spinel, a kind of alkaline metal, appearing on the top surface might improve the formation of $CoSiO_3/Co(AlO_2)_2$ for metal-ceramic bonding [54,55]. On the other hand, chromium is an acidic metal and chemical reaction between Cr and Si/Al is different. In addition, the enhanced roughness of spinel intrusion can provide favourable conditions for chemical bonding with more places for the reaction. From microstructural images, it looks that the intrusion is relatively regular, preventing defects during ceramic penetration. The regular shape of intrusion together with its chemical composition guaranteed the mechanical interlocking and ceramic-metal bonding.

5. Conclusion

In this work, fully dense CoCrWCuSi alloys doped with 0 %, 0.5 %, 1.1 % B were manufactured by 3D printing, and pre-oxidation in simulated porcelain sintering condition was implemented, to investigate the influence of minor B alloying at physical and mechanical properties of the oxide scale.

Subsurface deformation zone was introduced with fine-grain, strain-induced martensitic transformation and γ-Co deformation twins, which plays a role of Cr supply path for a dense and continuous chromia growth under short exposure in high temperature air. The addition of B to CoCrWCuSi alloys slowed down the growth rate of chromia scale, leading to reduced growth stresses in the scale which is less susceptible ro mechanical damage. Meanwhile, boron tends to increase oxide scale's adhesion to the substrate from scratch tests.

Boron was segregated in the inner layer of the oxide scale, closely next to the scale-alloy interface, and overlaped with the SiO_2. With the addition of boron, the oxide scale changed from outer Cr_2O_3 and inner amorphous SiO_2 to sandwiched triple structure (outer $CoCr_2O_4$ spinel, middle Cr_2O_3 and inner SiO_2+B_2O_3). The synergistic effect of SiO_2 and B_2O_3 formation on improving oxidation resistance and oxide scale adhesion is probably due to this layer plays as an effective barrier for the transportation of reactants, slowing the oxide scale growth, thereby reducing its susceptibility to fracture. In addition, the pining like arrangement of this amorphous layer increased the actual interface area between the oxide scale and the matrix, and relieved the spallation susceptibility.

Due to instrumental limitation, boron at atomic level was not revealed in this work and it is the scope of next work to better understand its effect on oxide scale and interfacial structure. Porcelain sintering will be carried out to optimize the pre-oxidation behavior.

CRediT authorship contribution statement

Chun Yu: Writing – original draft, Investigation, Funding acquisition, Formal analysis, Data curation. **Liangzhen Zhu:** Writing – original draft, Investigation, Formal analysis, Data curation. **Yanjin Lu:** Project administration, Data curation, Conceptualization. **Yi Zhao:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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